Question 1

Mouhamed Cisse

2026-02-10

STAT 4C03

```
shuttle = read.csv("Shuttle.csv")
```

d)

```
# The log-likelihood gradient function
gradient <- function(beta, x, y){
  beta0 <- beta[1]
  beta1 <- beta[2]

  eta <- beta0 + beta1 * x
  pi  <- pnorm(eta)      # normale CDF function
  phi <- dnorm(eta)      # normale density function

  grad0 <- sum(phi * (y - pi) / (pi * (1 - pi)))
  grad1 <- sum(x * phi * (y - pi) / (pi * (1 - pi)))

  c(grad0, grad1)
}

# The log-likelihood hessian function
hess <- function(beta, x, y){
  eta <- beta[1] + beta[2] * x
  pi  <- pnorm(eta)
  phi <- dnorm(eta)

  w <- (phi^2)/(pi*(1-pi)) # + (y-pi)*phi*eta/(pi*(1-pi)) By adding this term the NR algorithm didn't c

  h11 <- -sum(w)
  h12 <- -sum(w * x)
  h22 <- -sum(w * x^2)

  matrix(c(h11, h12,
           h12, h22),
         nrow = 2, byrow = TRUE)
}
```

```r
# Newton-Raphson algorithm for probit link
Newton.Raphson <- function(beta.init, x, y, max.its = 10){

  beta <- beta.init
  lld<- c(numeric(max.its), numeric(max.its))

  # Norm of the score at initial value
  lld [1] <- c(gradient(beta0, x, y)[1], gradient(beta0, x, y)[2])

  for(i in 2:max.its+1){

    grad <- gradient(beta, x, y)
    H <- hess(beta, x, y)

    beta.new <- as.vector(beta - solve(H) %*% grad)

    lld <- c(gradient(beta.new, x, y)[1], gradient(beta.new, x, y)[2])

    beta <- beta.new

    cat("Iteration", i-1,
    ": betahat.0 =", beta[1],
    "betahat.1 =", beta[2],
    "log-likelihood derivative at beta.0 =", lld[1],
    "at beta.1 =", lld[2],
    "\n \n")
  }

  cat("<----------------------------------------------------------------------------------------------------
  beta
}
```

```r
# application
x = shuttle$temp
y = shuttle$damage
beta0 = qnorm(mean(y))
beta.init = c(beta0, 0)
betahat <- Newton.Raphson(beta.init, x, y)
```

```
## Warning in lld[1] <- c(gradient(beta0, x, y)[1], gradient(beta0, x, y)[2]):
## number of items to replace is not a multiple of replacement length

## Iteration 2 : betahat.0 = 6.918976 betahat.1 = -0.1068194 log-likelihood derivative at beta.0 = -0.94
##
## Iteration 3 : betahat.0 = 8.413364 betahat.1 = -0.1298097 log-likelihood derivative at beta.0 = -0.0
##
## Iteration 4 : betahat.0 = 8.740499 betahat.1 = -0.134624 log-likelihood derivative at beta.0 = 0.019
##
## Iteration 5 : betahat.0 = 8.770736 betahat.1 = -0.1350378 log-likelihood derivative at beta.0 = 0.00
##
## Iteration 6 : betahat.0 = 8.774496 betahat.1 = -0.1350901 log-likelihood derivative at beta.0 = 0.000
##
## Iteration 7 : betahat.0 = 8.774902 betahat.1 = -0.1350957 log-likelihood derivative at beta.0 = 2.448
```

```
##
## Iteration 8 : betahat.0 = 8.774949 betahat.1 = -0.1350964 log-likelihood derivative at beta.0 = 2.790
##
## Iteration 9 : betahat.0 = 8.774954 betahat.1 = -0.1350965 log-likelihood derivative at beta.0 = 3.104
##
## Iteration 10 : betahat.0 = 8.774954 betahat.1 = -0.1350965 log-likelihood derivative at beta.0 = 3.49
##
## <-------------------------------------------------------------------------------------------------
##  betahat.final = ( 8.774954 , -0.1350965 ) Final loglikelihood derivatives value 3.490742e-08 at beta
```

e)

```
# compitation of the asymptotic standard errors of the estimators.
I_obs  <- -hess(betahat, x, y) # fisher information
VarHat <- solve(I_obs) # variance of betahat under normality assumption
SE     <- sqrt(diag(VarHat)) # standard errors computation for beta 0 and beta 1 respectively

SE
```

```
## [1] 3.87244728 0.05646599
```

The error is larger for beta 0 compared to beta 1.